

Name: _____

Translations. Find each image or rule.

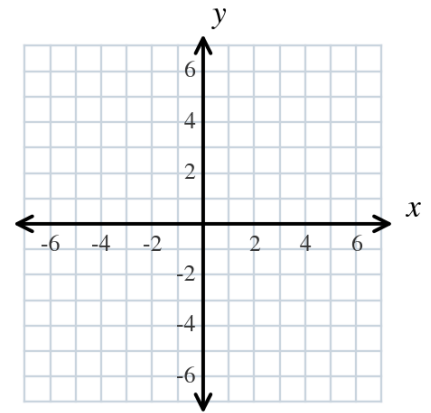
1. Find the image of $A(-6, 3)$ after a translation along $\langle 5, -8 \rangle$.
2. Find the component form of the vector that translates $B(4, -2)$ to $B'(-3, 6)$.
3. Use the rule $(x, y) \rightarrow (x + 2, y - 9)$. What is the image of $C(-1, 4)$?
4. Use the same rule. What is the preimage of $D'(6, -2)$?
5. A figure is translated along $\langle 7, -4 \rangle$ and then along $\langle -9, 6 \rangle$. Write ONE rule.

Use the coordinate plane at the right.

6. Graph $\triangle EFG$ with $E(-5, 2)$, $F(-2, 4)$, and $G(-1, 0)$ and its image after the translation $(x, y) \rightarrow (x + 6, y - 5)$. List the image coordinates.

Reflections. Find each image.

7. $H(2, -7)$ in the x -axis
8. $J(-5, 1)$ in the y -axis
9. $K(3, 8)$ in the line $y = x$
10. $L(-4, 6)$ in the line $y = -x$



Reflections in horizontal and vertical lines.

- 11.** Find the image of $M(7, -3)$ after a reflection in the line $x = 2$.
- 12.** Find the image of $N(-1, 5)$ after a reflection in the line $y = -1$.

Rotations about the origin. Find each image.

- 13.** $P(-2, 6)$ after a 90° rotation
- 14.** $Q(4, -5)$ after a 180° rotation
- 15.** $R(-3, -1)$ after a 270° rotation

Use the second coordinate plane at the right.

- 16.** Graph $\triangle STU$ with $S(1, 3)$, $T(4, 5)$, and $U(5, 1)$ and its image after a 90° rotation about the origin. List the image coordinates.

Compositions and symmetry.

- 17.** Find the final image of $\overline{VW}$ with $V(3, -4)$ and $W(6, -1)$ after a reflection in the y -axis followed by a 180° rotation about the origin.
- 18.** Find the final image of $\overline{XY}$ with $X(-5, 2)$ and $Y(-2, 4)$ after the translation $(x, y) \rightarrow (x + 4, y - 7)$ followed by a 90° rotation about the origin.
- 19.** How many lines of symmetry does a regular hexagon have? Describe every rotation of 180° or less that maps it onto itself.
- 20.** Explain why the image of any figure under a translation, a reflection, or a rotation is always congruent to the preimage.

